Annex I. Debt Sustainability Analysis

The debt stock was 71 percent of GDP at end-2017, well above prudent levels for El Salvador.[1] The debt ratio is higher by about 7 percentage points due to the recent one-off downward revision in nominal GDP. Some liabilities (notably pension recognition bonds) are not included in the estimated stock. Reflecting deficit reduction in 2016-17 and a recent pension reform, the debt ratio is projected to almost stabilize over the medium term, in contrast to the pronounced upward debt dynamics projected during the 2016 Article IV Consultation. The gross public financing needs do not look high compared to other emerging markets, but are elevated for a fully dollarized economy.[2] Eurobond repayments in 2019, 2023, and beyond need to be carefully managed. Susceptibility to growth and financing shocks are key risks. Vulnerabilities are moderated by a long maturity of existing debt, a stable investor base, and low implicit pension debt due to the ongoing transition to a defined-contribution system.

A. Key Assumptions

1. Debt definition. As in previous consultations, El Salvador's public debt is defined as the sum of gross debt of the nonfinancial public sector (NFPS) and the external debt of the central bank. The data on NFPS is reported by the Ministry of Finance. In staff's definition, the NFPS includes pension-related debt (formerly called *CIP-A* bonds that were issued to finance current public pension payments). This broad definition better captures quasi-fiscal liabilities related to public enterprises and the public financial sector. However, the debt definition excludes pension system's recognition bonds (e.g., *CIP-B* bonds, amounting to about 7 percent of GDP),[3] municipal debt (1½ percent of GDP), and some public enterprise debt.[4]

2. Macroeconomic outlook. The **baseline scenario** reflects the estimated growth potential of 2.2 percent, with somewhat higher growth in 2018-2019 due to expected U.S. economy's recovery. The fiscal projection incorporates the estimated structural component of the deficit reduction observed in 2016-17 and the effects of the 2018 budget, including the September 2017 pension reform. It is assumed that there is no further fiscal adjustment, with the primary balance gradually improving reflecting continued favorable fiscal effects from the pension reform. Reflecting higher interest bill from rising interest rates, the headline overall deficit would increase mildly over the medium term, exceeding 3 percent of GDP by 2023. The growth/interest differential would remain negative. The scenario assumes that financing gaps are filled with long-term loans from private creditors. Inflation is expected to remain anchored by the fully dollarized regime.

[1] Staff estimates public debt threshold for El Salvador within the 40–50 percent range (see IMF Country report No 15/13).

[2] In the absence of a fully-fledged LOLR role of the central bank, fiscal gross financing requirements in dollarized economies have been significantly lower than in other economies.

[3] The recognition bonds have been issued to validate the rights of pensioners who had contributed to the defined benefit system but would be receiving pension benefits from the defined contribution system.

[4] The exact amount of under-recorded debt would be determined as part of the ongoing reconciliation exercise between BCR and MoF.

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An alternative fiscal **adjustment scenario** envisions a cumulative fiscal consolidation of 2 percent of GDP by 2020. Under plausible multiplier assumptions (0.7 on impact and 0.3 in the longer-term), this effort would bend the public debt curve and contribute to a sustained reduction in the debt ratio below 60 percent of GDP by 2024-25 and below a more comfortable level of 50 percent of GDP by 2030.

Fiscal Indicators in an Adjustment Scenario (percent of GDP) 1/

	2018	2019	2020	2021	2022	2023	2024	2025
GDP growth (percent)	2.3	1.6	1.4	2.4	2.4	2.4	2.3	2.3
Overall balance	-2.2	-1.8	-0.7	-0.6	-0.6	-0.6	-0.7	-0.7
Primary balance	1.4	2.0	3.2	3.4	3.5	3.5	3.5	3.5
Gross financing need	6.7	8.8	4.1	4.5	4.5	6.4	n.a.	n.a.
Public debt	70.1	69.5	67.9	65.8	63.6	61.7	59.8	58.0

Source: Fund staff estimates.

1/ Consolidation measures of 1 percent of GDP in 2019 and 1.3 percent of GDP in 2020, with multipliers of 0.7 short term and 0.3 long-term, followed by an increase in capital spending by 0.2 percentage point starting from 2021.

B. Results and Assessment

3. Baseline scenario results. The public debt ratio would remain around 70 percent of GDP in 2023, being on a mildly upward trajectory thereafter. The gross financing needs would average 7½ percent of GDP over 2018-23, increasing to 9¾ percent of GDP in 2019 due to a Eurobond repayment that year. There is clearly **no fiscal space** with the debt dynamics rising and output gap near zero.

4. Drivers. The debt dynamics are primarily driven by real interest rates, contributing about 3 percent of GDP to growth in the ratio on an annual basis, while primary surpluses and real GDP growth would reduce the ratio by about 1½ percent of GDP annually each.

5. Stress tests. Mostly standard shock scenarios have been considered for comparability, except the natural disaster impact, which assumes a decline of GDP growth by 3 percentage points the first year and 2 percentage point the following year. Real GDP and contingent liability shocks have a relatively significant impact, with the latter particularly costly since it significantly increases financing needs. A combined macro-fiscal shock and a natural disaster shock would be also challenging.

6. Assessment. Standard debt profile characteristics are mixed (see heat-map). The heat-map may understate some risks, including because: (i) the gross financing threshold of the heat map (15 percent of GDP) does not account for funding constraints of fully dollarized economies (their gross financing requirements tend to be substantially smaller than in other emerging economies); and (ii) the measured share of "foreign currency" debt (close to zero) reflects the legal adoption of the U.S. dollar, not the implied benefits of issuing own-currency liabilities.

7. Mitigating factors. Unlike many countries, El Salvador on current policies has effectively pre-funded some of its future pension liabilities due to its ongoing transition toward a fully self-

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financed defined-contribution system. Existing debt has a relatively long average debt maturity and a stable investor base (over one-half of the debt is held by domestic pension funds and official creditors) – in the latter aspect, the heat-map may overstate risks from "nonresident holdings."

8. El Salvador's **pension system** would be an additional driver of long-term debt. The September 2017 pension reform has a substantial favorable effect on the medium-term public debt dynamics, but creates a long-term liability that could become significant after 2040. While, the main actuarial study commissioned by the government projects the latter effect to be moderate and not altering the declining pension spending to GDP ratio through 2100, it hinges on assumptions that may not fully account for long-term population aging.[5] Furthermore, without parametric reforms, "social sustainability" of pensions is a chronic problem as replacement rates and coverage would remain very low.[6]

9. **Idiosyncratic risks and issues.** These include: (i) risks of potential reversal of the recent deficit reduction, particularly due to potential political pressures for higher spending in the 2018-19 electoral season; (ii) risks of potentially fiscally-costly adjustments to the pension system in response to a possible fall-out from the system's social sustainability problems, although these risks have been mitigated with the recent reform; (iii) potential assumption of enterprise debt; (iv) financing challenges due to potential difficulties in assembling a two-thirds majority in Congress that is required to authorize long-term borrowing. This would be important to address in the context of 2019 Eurobond repayment and similar payments in 2023-24.

[5] El Salvador is so far in a demographic "sweet spot," but the share of population over 64 is projected to increase from 7 percent now to 16 percent in 2050 and 29 percent in 2100 (ECLAC estimates).

[6] See IMF Country report 16/209.

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Figure 1. El Salvador: Public Sector Debt Sustainability Analysis Baseline Scenario

(In percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

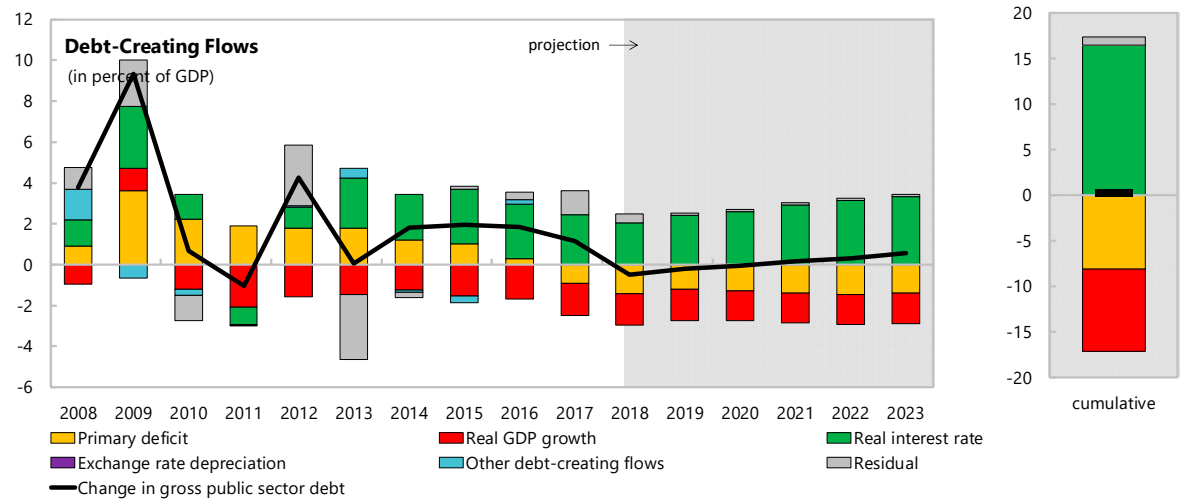
	Actual			Projections					
	2008-2015 [2/]	2016	2017	2018	2019	2020	2021	2022	2023
Nominal gross public debt	59.8	69.5	70.6	70.1	69.9	69.9	70.0	70.3	70.9
Public gross financing needs	8.9	8.5	8.8	6.7	9.7	6.2	6.8	7.0	9.1
Real GDP growth (in percent)	1.9	2.6	2.3	2.3	2.3	2.2	2.2	2.2	2.2
Inflation (GDP deflator, in percent)	2.0	0.6	1.4	2.3	2.2	2.2	1.9	2.0	1.9
Nominal GDP growth (in percent)	4.0	3.2	3.7	4.7	4.6	4.4	4.2	4.2	4.2
Effective interest rate (in percent) [4/]	4.9	4.7	5.1	5.4	5.8	6.1	6.3	6.7	6.9

As of April 05, 2018

Sovereign Spreads		
EMBIG (bp) 3/		369
5Y CDS (bp)		n.a.
Ratings	Foreign	Local
Moody's	B3	B3
S&Ps	CCC	CCC
Fitch	B-	B-

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2015	2016	2017	2018	2019	2020	2021	2022	2023	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	2.6	1.9	1.1	-0.5	-0.2	0.0	0.2	0.3	0.6	0.3	
Identified debt-creating flows	2.4	1.5	0.0	-0.9	-0.3	-0.2	0.1	0.2	0.5	-0.7	
Primary deficit	1.8	0.3	-0.9	-1.4	-1.2	-1.3	-1.4	-1.4	-1.4	-8.1	1.8
Primary (noninterest) revenue and grants	20.8	22.0	22.6	22.9	22.7	22.4	22.3	22.3	22.2	134.9	
Primary (noninterest) expenditure	22.6	22.3	21.7	21.5	21.5	21.1	20.9	20.9	20.9	126.8	
Automatic debt dynamics [5/]	0.5	1.0	0.9	0.5	0.9	1.1	1.5	1.7	1.8	7.4	
Interest rate/growth differential [6/]	0.5	1.0	0.9	0.5	0.9	1.1	1.5	1.7	1.8	7.4	
Of which: real interest rate	1.6	2.7	2.4	2.0	2.4	2.6	2.9	3.1	3.3	16.5	
Of which: real GDP growth	-1.1	-1.7	-1.6	-1.6	-1.5	-1.5	-1.5	-1.5	-1.5	-9.0	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Privatization/Drawdown of Deposits	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Increase BCR's external debt	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	0.2	0.4	1.2	0.4	0.1	0.1	0.1	0.1	0.1	1.0	



Source: Fund staff estimates and projections.

1/ Public sector is defined as non-financial public sector, including external central bank debt.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

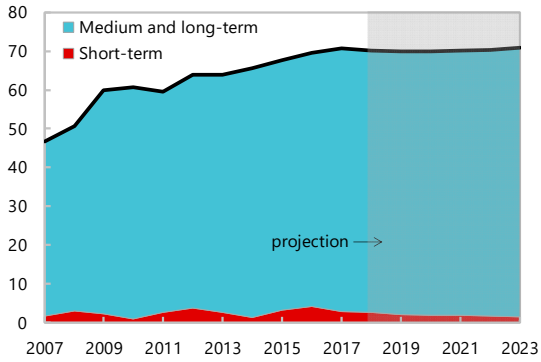
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Figure 2. El Salvador: Public DSA – Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

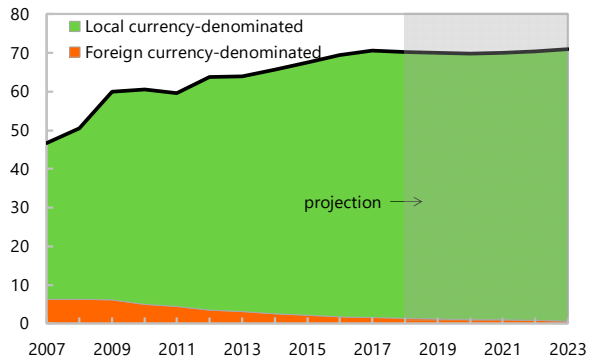
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)



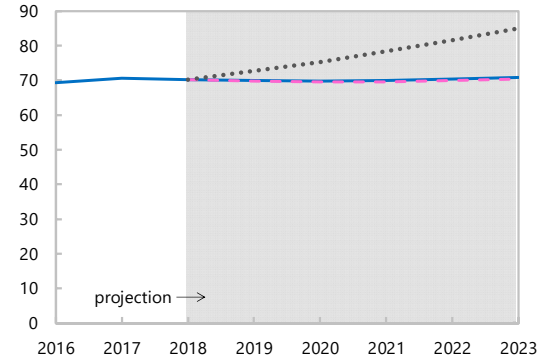
Alternative Scenarios

Historical

Constant Primary Balance

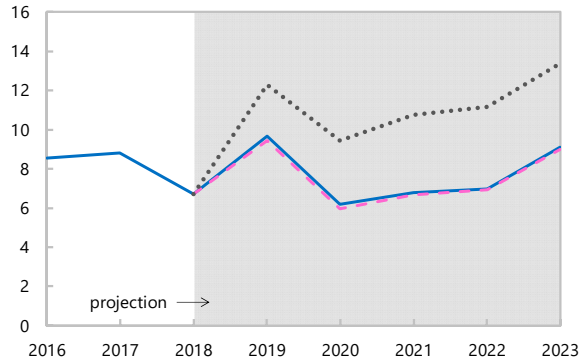
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	2.3	2.3	2.2	2.2	2.2	2.2
Inflation	2.3	2.2	2.2	1.9	2.0	1.9
Primary Balance	1.4	1.2	1.3	1.4	1.4	1.4
Effective interest rate	5.4	5.8	6.1	6.3	6.7	6.9
Constant Primary Balance Scenario						
Real GDP growth	2.3	2.3	2.2	2.2	2.2	2.2
Inflation	2.3	2.2	2.2	1.9	2.0	1.9
Primary Balance	1.4	1.4	1.4	1.4	1.4	1.4
Effective interest rate	5.4	5.8	6.1	6.3	6.7	6.9

Historical Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	2.3	2.0	2.0	2.0	2.0	2.0
Inflation	2.3	2.2	2.2	1.9	2.0	1.9
Primary Balance	1.4	-1.4	-1.4	-1.4	-1.4	-1.4
Effective interest rate	5.4	5.8	6.0	6.1	6.4	6.5

Source: Fund staff estimates and projections.

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